

Practical tasks

— bearings and scale drawings

The national “practical tasks” lesson, carrying the Cambridge Stage 9 topic of bearings.

LESSON 31

1 × 40 MIN

GEOMETRY 8, TEMA 28

STAGE 9 · 13.1 [CAMBRIDGE INSERT]

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Measure and write a bearing as three figures from north, clockwise.
- Find a back bearing.
- Make and read a scale drawing.
- Combine bearings with right-angle trigonometry to find a distance.

TEXTBOOK

National	Tema 28, pp. 64–66
Cambridge	Learner’s Book pp. 271–277
Workbook	Workbook 13.1

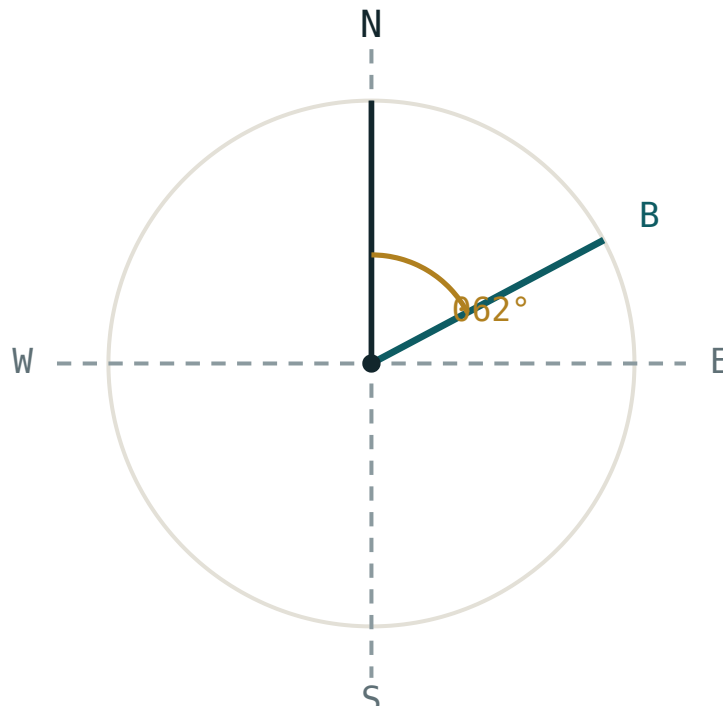
01

Explanation

What a bearing is

DEFINITION

A **bearing** is an angle measured **from north, clockwise**, and written with **three figures**.



The bearing of B from A is 062° — measured from north, turning clockwise.

All three parts matter. 62° must be written 062° ; a bearing measured anticlockwise or from another direction is simply wrong. The four cardinal directions are 000° (N), 090° (E), 180° (S) and 270° (W).

“THE BEARING OF B FROM A”

Stand at A, face north, and turn clockwise until you are looking at B. The point you stand at is the one after the word “from”.

Back bearings

The bearing of A from B is the **back bearing**. Because the two north lines are parallel, the two bearings differ by exactly 180° :

$$\text{back bearing} = \text{bearing} + 180^\circ \quad (\text{or } - 180^\circ \text{ if that exceeds } 360^\circ)$$

Bearing $062^\circ \rightarrow$ back bearing 242° . Bearing $310^\circ \rightarrow$ back bearing 130° .

Scale drawings

A scale of $1 : 50\,000$ means 1 cm on the paper is 50 000 cm — that is 500 m — on the ground. To make a scale drawing:

1. Draw a north line at the starting point.
2. Measure the bearing with a protractor, clockwise from north.

3. Convert the real distance to a paper distance and mark it off.

4. Repeat from the new point, with a fresh north line.

Then measure the answer off the drawing and convert back. A scale drawing gives a good enough answer for most practical work — and it is a check on the trigonometry.

Bearings with trigonometry

When the bearings are 090° apart, the path makes a right angle and the whole of this chapter applies.

Example. A ship sails 8 km on a bearing of 000° then 6 km on 090° . The two legs are perpendicular, so the distance from the start is $\sqrt{64 + 36} = 10$ km, and its bearing from the start is $\tan^{-1} \frac{6}{8} \approx 037^\circ$.

02 Worked examples

EXAMPLE 1 The bearing of B from A is 145° . Find the bearing of A from B.

① $145^\circ + 180^\circ = 325^\circ$
Add 180° for the back bearing.

② $325^\circ < 360^\circ$
so no adjustment is needed.

③ 325°

ANSWER

325°

EXAMPLE 2 A ship sails 12 km due north, then 5 km due east. Find its distance and bearing from the start.

① North then east — the two legs are perpendicular.

② $d = \sqrt{144 + 25} = 13 \text{ km}$
Pythagoras.

③ $\tan \alpha = \frac{5}{12} \approx 0.4167$
East over north.

④ $\alpha \approx 22.6^\circ$, so the bearing is 023°
Measured clockwise from north.

ANSWER

13 km on a bearing of 023°

EXAMPLE 3 On a map of scale 1 : 25 000 two towns are 7.2 cm apart. Find the real distance.

① $1 \text{ cm} \rightarrow 25\,000 \text{ cm}$
The scale.

② $25\,000 \text{ cm} = 250 \text{ m}$
Convert to metres.

③ $7.2 \cdot 250 = 1800 \text{ m}$

④ $= 1.8 \text{ km}$

ANSWER

1.8 km

03 Interactive model

Have a learner walk a bearing across the classroom while the class calls out the angle.

Bearings, step by step

Bearing 062° .

1 The two north lines are parallel, so the bearings differ by 180° .

2 $062^\circ + 180^\circ = 242^\circ$

3 Under 360° , so no adjustment.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bearing	Azimut	Азимут
North	Shimol	Север
Clockwise	Soat yo'nalishi bo'yicha	По часовой стрелке
Back bearing	Teskari azimut	Обратный азимут
Scale	Masshtab	Масштаб
Scale drawing	Masshtabli chizma	Чертёж в масштабе
Compass	Kompas	Компас
Due east	Aniq sharq	Точно на восток
Protractor	Transportir	Транспортир

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A bearing is measured:

from east, anticlockwise

from north, clockwise

from north, anticlockwise

from the horizontal

Due west as a bearing is:

090°

180°

270°

360°

The bearing of B from A is 100°. The bearing of A from B is:

080°

200°

260°

280°

On a 1 : 50 000 map, 1 cm represents:

50 m

500 m

5 km

50 km

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Write due north as a bearing.

ANSWER 000°

2. Write due east as a bearing.

ANSWER 090°

3. Write due south as a bearing.

ANSWER 180°

4. Write due west as a bearing.

ANSWER 270°

5. Write 45° north-east as a three-figure bearing.

ANSWER 045°

6. The bearing of B from A is 070° . Find the back bearing.

ANSWER 250°

7. On a $1 : 100\,000$ map, 1 cm represents how far?

ANSWER 1 km

Medium · the standard question

7 PROBLEMS

1. *The bearing of B from A is 145° . Find the bearing of A from B.*

ANSWER 325°

2. *The bearing of B from A is 310° . Find the back bearing.*

ANSWER 130°

3. *On a 1 : 25 000 map two towns are 7.2 cm apart. Find the real distance.*

ANSWER 1.8 km

4. *A ship sails 12 km north then 5 km east. Find its distance from the start.*

ANSWER 13 km

5. *The same ship: find its bearing from the start.*

ANSWER 023°

6. *A real distance of 4 km on a 1 : 50 000 map is how many cm?*

ANSWER 8 cm

7. *The bearing of B from A is 205° . Find the back bearing.*

ANSWER 025°

Hard · stretch and reason

7 PROBLEMS

1. A ship sails 9 km on 000° , then 12 km on 090° . Find its distance and bearing from the start.

ANSWER 15 km on 053°

2. A walker goes 5 km on 090° then 5 km on 180° . Find the distance and bearing from the start.

ANSWER $5\sqrt{2} \approx 7.07$ km on 135°

3. The bearing of B from A is 062° and $AB = 20$ km. How far north and how far east is B?

ANSWER $20 \cos 62^\circ \approx 9.4$ km north, $20 \sin 62^\circ \approx 17.7$ km east

4. A plane flies 100 km on a bearing of 030° . Find its change in latitude (north) and longitude (east) in km.

ANSWER 86.6 km north, 50 km east

5. Two ports are 40 km apart, one due east of the other. A ship is 30 km north of the western port. Find its bearing from the eastern port.

ANSWER $\tan \alpha = \frac{40}{30}$, so the bearing is $360^\circ - 53.1^\circ = 307^\circ$.

6. Explain why a back bearing is exactly 180° from the original.

ANSWER The two north lines are parallel, so the bearings are co-interior angles adding to a straight-line turn.

7. A map has scale 1 : 20 000. A lake measures 3 cm by 4 cm on it. Find its real area in km^2 .

ANSWER $600 \text{ m} \times 800 \text{ m} = 0.48 \text{ km}^2$

07

Homework**Homework — 5 problems**

Geometry 8, Tema 28, pp. 64–66, and Cambridge Learner's Book 13.1. Bring a protractor.

- Write these as three-figure bearings: north-east, south-west, due south.
- The bearing of B from A is 118° . Find the bearing of A from B.
- On a 1 : 50 000 map two villages are 5.6 cm apart. Find the real distance.

4. *A ship sails 8 km north then 15 km east. Find its distance and bearing from the start.*
5. *Make a scale drawing (1 cm : 1 km) of a journey 6 km on 070° followed by 4 km on 160° , and measure the direct distance home.*